

Electromagnetism Learning Plan

Learning-Note

7 October 2026

Learning method

Build physical meaning before manipulating formulas. For each concept, identify what can be observed or measured, state the mathematical definition, and distinguish it from a physical law or theorem. Stay within the requested concept and advance only after the learner confirms understanding. A planned subject is not a completed lesson.

Current state

- Flux, divergence, and curl: the learner has confirmed understanding of their definitions and their general purpose. The first note records physical interpretations, applications, and related laws.
- Electric and magnetic fields: the supplied side-chat summary has been checked and incorporated with assumptions about probes, magnetic dipoles, spin, and energy.
- Impedance and voltage/current reflection formulas: previously discussed and reported mathematically understood. Their physical interpretation still needs development.
- Laws and experiments: a separate timeline note now records Coulomb, steady-current magnetism, Gauss, Faraday, Lenz, Stokes, Maxwell's extension, and the force law. It includes the missing EMF and independent-measurement explanations. Recording a lesson does not establish confirmed understanding.
- **Primary learning direction, from 7 October 2026:** Follow Edward M. Purcell and David J. Morin, *Electricity and Magnetism*, 3rd edition (2013), as the main sequence. The existing notes support the book-led study; the next subjects follow the book rather than a direct jump to Maxwell's equations.
- **Reading checkpoint:** Chapter 1, *Electrostatics: charges and fields*. Recent discussion covers Coulomb's law and superposition, Gauss's law, spherical/line/sheet charge distributions, forces on a charge layer, and field energy. Problem 1.33 was worked through to derive the two-charge interaction energy from the field integral. Discussion and a worked solution do not imply confirmed mastery.

Proposed sequence

1. **Book-led foundations through Maxwell's equations.** Continue Chapter 1, then study potential and conductors in Chapters 2–3. Work selected problems to check derivations. Follow the book toward magnetism, induction, Maxwell's equations, and waves, explaining experimental assumptions and mathematical steps before advancing.
2. **Displacement current and charge conservation.** Examine a charging capacitor, compare surfaces sharing a boundary, and explain why the time-dependent electric-field term is needed.
3. **Fields at conductors and energy flow.** Introduce surface charge, surface current, ideal-conductor boundary conditions, the Poynting vector, and field energy with explicit assumptions.
4. **Transmission lines as guided fields.** Connect a TEM line's transverse electric field to voltage, magnetic circulation to current, and field storage to distributed capacitance and inductance.
5. **Reflection at an ideal short and open.** Fix current and voltage reference directions. Explain how the conductor responds, how fields superpose, and how backward energy propagation can coexist with a reflected current of the same sign. Distinguish incident, reflected, and total quantities.
6. **Apply the picture to the AN720 antenna.** First identify the exact model, feed, conductor connections, substrate, and relevant geometry using verified documentation and photographs. Then connect current/charge distributions, resonance, matching, polarization, and radiation to the confirmed structure.

Repository organization and updates

Each subject has its own folder. The current folder is `Electromagnetism-Foundations/`, containing two independent notes:

- `fields-and-vector-calculus.tex` and its PDF.
- `laws-experiments-and-timeline.tex` and its PDF.

Repository documents are the source/PDF pairs for `Documents/plan.tex` and `Documents/changes.tex`.

For the next lesson, extend the foundations note when material directly develops it; create a separate topic folder when the subject merits an independent note. Append a dated change entry and regenerate affected PDFs. Use stable filenames without release numbers or duplicate revision folders.

Only `.tex` and `.pdf` files are intended for Git. Auxiliary outputs and unrelated files are excluded using a repository-local Git exclude rule. Git configuration is not part of an Overleaf upload. Each document uses ordinary pdfLaTeX packages and compiles independently; no external bibliography is needed.

Suggested books and reading path

These are recommendations based on the learner's preference for experimental and physical meaning before algebra. Coverage is checked against publisher information; suitability and sequence are editorial judgments.

1. **Current primary text: Purcell and Morin, Electricity and Magnetism, 3rd edition (2013).** Reading began with Chapter 1: charges, fields, flux, and Gauss. Chapters 2–3 develop potential and conductors; Chapters 5–6 cover moving charges and magnetism; Chapter 7 covers induction; Chapter 9 covers Maxwell and waves. Chapters 10–11 address fields in matter and magnetic moments. This text supplies the physical foundations for later transmission-line and antenna study.
Cambridge publisher page.
2. **For systematic mathematical derivations: Griffiths, Introduction to Electrodynamics, 5th edition.** Chapter 1 covers vector analysis; Chapters 2–7 develop electrostatics through electrodynamics; later chapters cover conservation, waves, fields, and radiation. Use as a rigorous companion when ready for more calculation. It is a physics foundation rather than an RF hardware design text.
Cambridge publisher page.
3. **Then connect fields to wireless hardware: Pozar, Microwave Engineering, 4th edition.** Prioritize Chapters 1–3 (fields, transmission lines, waveguides), Chapter 4 (network analysis), and Chapter 5 (matching). It directly supports the planned impedance/reflection and RF work, but its initial EM review is not a substitute for the foundational explanations requested here.
Wiley companion contents.
4. **For antenna analysis and design: Balanis, Antenna Theory: Analysis and Design, 4th edition.** Use after field propagation and transmission-line basics. Its radiation, antenna parameters, impedance, polarization, and antenna-geometry treatment is relevant to later AN720 analysis. Identify the antenna's actual structure before selecting a model.
Wiley antenna-text companion site.

Free supplementary reading: The Feynman Lectures, Volume II, especially Chapters 2–3 for vector fields and Chapters 16–18 for induction and Maxwell. They are useful for physical explanations, alongside a structured textbook and exercises.

https://www.feynmanlectures.caltech.edu/II_toc.html

A practical sequence is Purcell–Morin for understanding, Pozar for guided fields and RF, then Balanis for antennas. Griffiths can supply more detailed mathematical practice in parallel. No single text replaces careful distinction between a definition, a mathematical identity, and an experimentally supported law.